

Lab I: deterministic filtering and linear stochastic models with applications to ECG and EEG

Lab sessions: September 2nd and September 9th.

General instructions

How to do the lab

This lab consists of a **pre-lab task** and **three lab tasks**. The pre-lab task is meant to serve as preparation and consists of reading the necessary sections of the course literature as well as theoretical questions.

You should work on the lab in pairs of two. Note that larger groups are not allowed.

Take the opportunity to attend the lab sessions. During these sessions, you can work on the problems and ask questions. You are recommended to do the labs using MATLAB, but you are welcome to use other languages as well.

Report grade

You are examined for this lab by submitting (on time, see deadline on MyCourses) a lab report. The report will be graded on the scale 0 – 3, where 0 means "fail".

- If your lab report receives the grade 0, you will be asked to re-submit an amended version. A re-submitted report can get at most grade 1.
- Note that in order to receive a passing grade on the course you need to get at least grade 1 for the reports of each of the three labs.

How to write a lab report and what to hand in

Your lab report is supposed to provide a full account of what you do, observe, reflect, and learn in the lab. It should thus contain descriptions of the signals and models that you use, describe the processing, and your reflections and conclusions. Provide relevant plots and figure, give them appropriate captions (i.e., descriptive figure texts) and legends, and refer to them in the text. In the lab description there are some plots requested already, but feel free (be creative and have fun!) to include additional ones.

In terms of writing style, the lab report should be a flowing text and not a random set of answers. In the lab description, you have several questions and encouragements to reflection. Weave these in seamlessly to your report. Think that the audience for your report is a student of this course or similar that has not read the lab description.

You should hand in:

- Your lab report (written in LaTeX or Word) as a single pdf-file. The report should be written in English. Even though you should write your lab report in pairs of two, you should hand in the report individually.
- On the report, write both your name and the name of your lab partner.
- The computer code that you have used to solve the problems of the lab and to generate the figures of your report. Hand this in as a single zip-file.

Rules on large language models, collaboration, and plagiarism

You may use large language models, such as ChatGPT, or other AI tools to aid your learning. However, using these tools to generate computer code or write your report is strictly forbidden and regarded as plagiarism or cheating, in addition to not being fun for either of us.

You are allowed, and encouraged, to discuss with the other students of the course, **but you should do the lab and write the report together with only your lab partner**, and the report should be written "in your own words". Use your common sense and avoid plagiarism.

Good luck and have fun!

Pre-lab preparation

These are theoretical tasks that should be completed **before** the lab. Note that you need these preparations in order to answer the questions of the lab and to write the lab report.

Preparation for ECG filtering

- a) Read Chapter 6, the introduction section of chapter 7, 7.1.1, and section 7.2 in Sörnmo and Lagunas.
- b) What are the relative pros and cons of using FIR and IIR filtering?
- c) How can the non-linearity of an IIR filter response be handled? Are there any limitations to using these techniques in practical applications?

Preparation for AR-modeling and EEG

- a) Read Chapter 2, 3.1, 3.4.1, 3.4.4, in Sörnmo and Lagunas, and sections 3.3 and 3.4 in Stoica and Moses.
- b) Write down the definition of an AR(p)-process and clearly state the assumptions on all the constituent parts.
- c) What is the covariance function for the noise component of the AR(p)-process?
- d) How are the parameters of an AR-process estimated using the Yule-Walker method? Give an explicit expression for the parameters in terms of the covariance function of the AR-process.
- e) Write down the definitions of the Akaike information criterion (AIC) and minimum description length (MDL) and describe how they are used in model order selection.
- f) Assume that you have data generated from an AR(2)-process to which you try to fit an AR(1)-model using the Yule-Walker method. Let the parameters of the AR(2)-process be a_1 and a_2 , and the parameter for the AR(1)-model be $\tilde{a}_1$. Find an expression for $\tilde{a}_1$ in terms of a_1 and a_2 .

Hint: what is the Yule-Walker expression for the estimate of the AR(1)-model and what is the covariance function for an AR(2)-model?

MATLAB practicalities

- ECG: Familiarize yourself with the Matlab functions `butterord`, `butter`, `firls`, `filter`, and `filtfilt`.
- AR/EEG: Familiarize yourself with the Matlab functions `xcov`, `iddata`, `arx`, `present`, and `filter`.

Lab task 1: Enhancing an ECG signal

In this task, you will enhance an ECG signal that has been corrupted by powerline interference, i.e., interference from the electric grid.

1. Download `ecg_data.mat` from the course home-page. This contains an ECG recording^a in the vector `y` together with a time-stamp vector `t`. The sampling frequency is $F_s = 500$ Hz.
2. Make a plot of the whole ECG signal as well as a zoomed-in plot of only the first second. What type of structures is it that you see? What are the different components called? (See Chapter 6.2.3).
3. Make a plot of the absolute value of the Discrete Fourier transform of the data (use the command `fft` with zero-padding). How does your picture compare to Figure 6.11 in the book?
4. How can you determine the fundamental frequency of the signal? (Hint: what is the relation between the frequencies of the harmonics for periodic signals?). What is the corresponding period of the signal? Given this, what is the subjects heart rate in beats/minute? How can you estimate the same information from the covariance function of the signal (use `covf`)?
5. Can you see the frequency component corresponding to the power grid? What frequency does it have?

IIR filtering

1. Design a **Butterworth IIR filter** for removing the power grid interference.
 - The filter should have a stop-band attenuation of 40 dB and a passband ripple of 3 dB.
 - What type of filter (high-pass, low-pass, band-pass,...) is suitable? (Hint: where do the relevant frequency components seem to be located relative to the power grid interference?)
 - What passband edge frequency and stop band edge frequency could be suitable?
 - Use the `butterord` function to compute the order of the Butterworth filter and corresponding cutoff frequencies. Note: the `butterord` requires the input frequencies to be normalized to between $[0,1]$, where 1 corresponds to π radians/sample. Thus, to convert a frequency f (in Hz) to the correct format, you write `w = f/fs * 2`, where `fs` is the sampling frequency.
 - Use the function `butter` with the output from `butterord` to compute the filter coefficients `[b,a]`.
 - Inspect your filter using the `freqz` command. Make a plot and comment on the magnitude and phase response.
2. Use your obtained Butterworth filter to filter the ECG signal using the command `filter`. Plot the result in a figure together with the original signal and zoom in on the first second. What do you observe? What is this caused by? (Hint: use your preparation tasks).
3. As an alternative, filter the signal using the command `filtfilt`. Plot the result in a figure together with the original signal and zoom in on the first second. What difference do you observe as compared to using `filter`? Why is this?

FIR filtering

1. Design an FIR with the same filter specifications (i.e., same type of filter with same edge frequencies) by the least-squares method using the `firls` command. You here have to select **the order N** (i.e., the length of the filter).
2. Inspect the magnitude and phase response of your filter using `freqz`. How big do you have to choose N to get a magnitude response similar to that of your IIR filter? Comment on the phase response: what difference do you observe compared to your IIR filter?
3. Filter the ECG signal with your obtained FIR filter using the `filter` command. Make a plot similar to the one you made using the IIR filter and comment on the differences. What are the differences caused by? What are the pros and cons of using IIR and FIR filters? What would you use in an application?

^aThe recording is the so-called MLII-lead, which is a modified version of lead II, i.e., corresponding to the right arm and left leg (see Figure 6.6 in Sörnmo and Lagunas). In MLII, the electrodes are placed on the chest instead of the limbs. A motivation for using this configuration is that it reduces movement artifacts and is more convenient to record.

Lab task 2: AR-model estimation

Fitting parametric models to data is a powerful way of extracting information from noisy measurements. In particular, using a model can often be more efficient in data usage as compared to non-parametric methods in the sense that considerably fewer measurements are needed to achieve the same quality of the information retrieved.

However, a model is only useful if it provides a reasonably accurate description of the data – using the “wrong” model may actually hurt you. An overly simplistic model may fail to capture important data characteristics. On the other hand, a more complicated model that fits the data well may have too many parameters to be accurately estimated from a finite number of measurements.

This task is concerned with estimating the parameters of **autoregressive (AR)** models from noisy data.

1. Download the file `ar_data.mat` from the course home-page. This contains a data vector `y`.
2. Compute and plot the first 50 lags of the covariance function of `y`, for example using the function `covf`. Does `y` look like a white noise or does there seem to be some structure?
3. For $p = 1, 2, \dots, 20$, fit $AR(p)$ -models to the data.
 - You can use this using the command `ar_model = arx(iddata(y),p)`. The estimated coefficients are retrieved as `ar_model.A`. Note that this vector also includes the leading coefficient 1.
4. For each p , save the variance of the prediction error and compute the MDL and the AIC. Make a plot of the three quantities as a function of p . Why are the curves different and how should they be interpreted?
5. What is the optimal $p = p_0$ according to MDL? Does AIC say the same thing?
6. Fit your optimal model according to `ar_model0 = arx(iddata(y),p0)`. You can get the estimated model coefficients together with confidence intervals using `present(ar_model0)`.
 - Are all coefficients statistically significantly different from zero?
 - Compute an estimate of the excitation noise for the $AR(p_0)$ -model (hint: use `filter`). Compute and plot the first 50 lags of the covariance function for the resulting process. Does it have the expected behavior?
7. Fit a “**too large**” model, i.e., test to fitting an $AR(p_0 + 1)$ -model to the data.
 - Are all coefficients statistically significantly different from zero? Are the coefficient values similar to those of the $AR(p)$ -model? What happened to the confidence intervals for the coefficients as compared to the $AR(p_0)$ -model? What are the practical consequences of this?
 - Compute an estimate of the excitation noise for the $AR(p_0 + 1)$ -model. Compute and plot the first 50 lags of the covariance function of the noise and comment on this in relation to the $AR(p_0)$ -model.
8. Fit a “**too small**” model, i.e., test to fitting an $AR(p_0 - 1)$ -model to the data.
 - Are all coefficients statistically significantly different from zero? Are the coefficient values similar to those of the $AR(p)$ -model? What happened to the confidence intervals for the coefficients as compared to the $AR(p_0)$ -model? What are the practical consequences of this?
 - Compute an estimate of the excitation noise for the $AR(p_0 - 1)$ -model. Compute and plot the first 50 lags of the covariance function of the noise and comment on this in relation to the $AR(p_0)$ -model.
9. If you concluded that $p_0 = 2$, comment on the parameter values of the $AR(p_0 - 1)$ -model in relation to your preparation task.
10. Now you will evaluate how well AIC and MDL works by using a simulation study.
 - Simulate $N = 500$ samples from the $AR(p_0)$ -model. This can be done using the commands `w = randn(N,1)`, `y = filter(1,ar_model0.A,w)`. Find the optimal model order p for this data according to AIC and MDL.
 - Repeat this in 100 simulations. In each simulation, record the optimal p according to AIC and MDL.
 - Make histograms of the distribution of optimal p according to AIC and MDL (hint: use `hist`).
 - What is your conclusion? Does AIC or MDL seem most reliable for doing model order selection?

Lab task 3: EEG-signal classification for brain-computer interface

In this task, you will look at data from a brain-computer interface (BCI) experiment. On the course home-page, you can find the file `bci_eeg.mat` containing two matrices `eeg_left_hand` and `eeg_right_foot`. Each column of the matrices corresponds to a 2-second EEG recording with sampling frequency 128 Hz, recorded using the same electrode.

For the data in `eeg_left_hand`, the subject has been asked to imagine moving their left hand, whereas for `eeg_right_foot` they have been asked to imagine moving their right foot. Your task is to investigate whether these signals are useful for a BCI, i.e., if it is possible to determine the desired movement from the EEG signal.

In particular, when fitting AR-models to the data, can we see a difference in the models for the "left hand" and "right foot" signals? In order to do this, we need to decide on a **good model order** for the AR-models, i.e., to select p in $\text{AR}(p)$. As to not compare apples and oranges, we want to use the same p for both "left hand" and "right foot" signals.

1. Download `bci_eeg.mat` from the course home-page and plot the signals for the "left hand" and "right foot".
2. Extract the first column from `eeg_left_hand`.
 - Fit $\text{AR}(p)$ -models for $p = 1, 2, \dots, 20$ to the data and compute the prediction error variance, the Akaike information criterion (AIC), and the minimum description length (MDL) for each model. Plot the three curves and comment on their behavior. What do the different curves mean? Why are they different and how should they be interpreted?
 - What is the optimal p according to MDL? Are all parameters statistically significant for this model? Does the covariance function of the prediction error behave as expected? Make plots to motivate your conclusions.

Repeat the same procedure for the first column of `eeg_right_foot` and comment on potential differences.

3. Fit $\text{AR}(p)$ -models for $p = 1, 2, \dots, 20$ for each column in the both data sets and report the optimal model orders according to MDL.
4. As we want to use the AR-models for classification, you now have to choose a single $p = p_0$ that should be use for all recorded signals. What model order would you use? Motivate your answer!
5. With your selected p_0 , fit $\text{AR}(p_0)$ -models for each of the signals.
 - In a single plot, plot the values of the estimated AR coefficients for the different signals and the two data sets. Clearly mark which coefficients belong to the "left hand" and "right foot" respectively (for example with different colors and markers).
 - Does it seems to be possible to distinguish between "left hand" and "right foot" signals based on the coefficient values? What coefficients seem to offer the most promising separation?
 - Comment on how you would use this in a BCI application.